

3D Object/Parts Detection in Manufacturing Environments

Dheenadayalan Tamilarasan
Muthuselvi
Responsible for model development,
and wrote sections 1 and 4

Sharik Anwar Zahir Hussain
Responsible for the front end and
maintaining the whole
documentation

Krishnapriya Krishnan
Santhadevi
Responsible for the GUI design and
wrote section 3

ABSTRACT

This paper focuses on developing a robust system for identifying the maturity levels of diverse manufacturing components. The system aims to improve productivity, reduce errors, and increase overall efficiency by utilizing a 2D camera and VGG Imagenet-based Convolutional Neural Network (CNN). The dataset consists of real-time images and synthetic 3D-CAD data, which is strategically collected to ensure representativeness and diversity, crucial for accurate classification across various manufacturing stages. Data augmentation techniques, including rotation and scaling, increase dataset heterogeneity and make the model robust against real-world variations. To be able to deploy the system in real time with limited resources, the project explores several CNN architectures while considering model complexity and computational efficiency into account. It focuses particular attention on the integration of the VGG Imagenet algorithm. Additionally, transfer learning plays a crucial role, as it leverages pre-trained knowledge from the VGG Net. Rigorous testing and validation using images of real production ensures that CNN performs under realistic conditions, including variations in lighting and field orientation. To ensure the model's reliability in practical scenarios, its ability to manage blind spots and partial views is evaluated. The developed system provides valuable insights into manufacturing processes after successful completion, enables data-driven decisions, and facilitates real-time operations. The system empowers manufacturers to implement targeted process improvements, reduce errors, and improve overall efficiency. The project contributes to the advancement of manufacturing systems but also highlights the incorporation of the VGG Imagenet algorithm, laying the groundwork for future innovations across material handling, process control, and overall manufacturing performance.

KEYWORDS

VGG Imagenet, Synthetic 3D-CAD Data, Defect Recognition, Transfer learning.

1 INTRODUCTION

In recent years, considerable research efforts have been directed towards advancing the automation of industrial quality control processes, with a specific emphasis on leveraging machine learning methodologies. Quality control, an essential aspect of industrial production, involves meticulous testing and measurement of products to ensure conformity to predefined specifications and standards. This paper contributes to the ongoing discourse by employing the VGG (Visual Geometry Group) neural network architecture, specifically the VGG Net, for automating quality control processes. Notably, transfer learning is applied, utilizing the pre-trained knowledge of the VGG Net on a comprehensive dataset, making it a valuable tool for applications with limited data availability.

In addition to the choice of model architecture, this research introduces a practical dimension by developing a simple Graphical User Interface (GUI). This GUI facilitates user-friendly interaction with the trained VGG Net, providing an intuitive tool for inspecting and analyzing manufacturing components. This research explores the applicability of VGG Net and transfer learning in automating quality control for this broader scope of industrial processes.

The specific objectives of this paper include:

- Employing the VGG Net for identifying and localizing defects in the dataset's defective parts through transfer learning.
- Investigating the effectiveness of transfer learning in adapting the model to the distinct characteristics of the dataset, showcasing its versatility across various manufacturing conditions.
- Introducing a simple GUI to enhance the usability and accessibility of the quality control system.

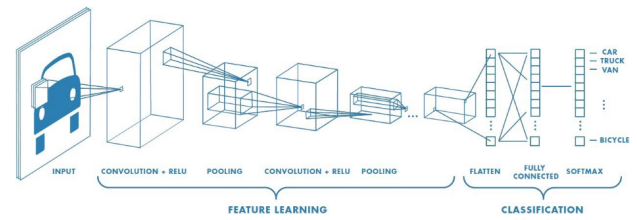


Figure 1: Work flow of VGG Net

The architecture shown in Figure 1 has been referred from GitHub and has the essential features of the VGG-16 model. Convolutional layer blocks are repeated and then a ReLU activation function is applied. The distances between these blocks are reduced by pooling layers. After the convolutional layers, there are several completely linked layers. A last classification layer made of softmax.

The paper's structure includes a comprehensive review of related work in Section 2, titled *State of the Art*, followed by the methodology and the analysis of the VGG Net architecture in Section 3, titled *Methodology*. Results obtained are presented in Section 5, with a discussion on the GUI development.

2 STATE OF THE ART

The field of defect detection in manufacturing parts has witnessed significant advancements driven by various methodologies and technologies. The manufacturing defects can arise from factors such as low metal temperature, cold dies, or insufficient venting. Attempting to get past these constraints, there has been an evolution

towards the use of machine learning (ML) to automate quality control procedures, eliminating the necessity for human intervention [3]. Numerous research endeavors have addressed the challenge of quality control in manufacturing parts. Training a machine vision system to analyze manufacturing components automatically could assist an industry to run more effectively and spend less money on operational costs [2].

An inspection system based on digital images examined using a hybrid processing algorithm was suggested to identify surface defects in casting products [4]. First, a genetic algorithm was employed to extract fundamental features utilizing structural techniques. As a result, the defects that were evaluated exhibited greater accuracy.

In [8], it was suggested to use a support vector machine (SVM) classifier to identify defects in images of highly reflected metals. In order to minimise noise, a wavelet transform was specifically performed on images of the products. The primary features in the spectral domain were then extracted after the digital images were transformed into binary based on the Otsu threshold. These features were given into the SVM model, which had an 85% success rate in classifying the associated defects.

Several machine learning algorithms were proposed in [5, 6] for the automated evaluation of casting surface quality. The part images in these particular articles were taken with a 3D-capable laser camera. Following that, certain features were taken out and trained using machine learning algorithms, including decision trees, SVM models, Bayesian networks, and the k-nearest neighbour algorithm.

Convolutional neural networks (CNNs) have been widely used in more recent studies for image processing because of their deep design, which automatically extracts key characteristics. In particular, an 11-layer CNN was used [7], which resulted in a 99% classification accuracy for components that have suffered damage. Showcasing the successful application of transfer learning to a small dataset, [1] highlights the potential for wider industrial adoption.

These state-of-the-art approaches showcase the evolution of techniques for defect detection, emphasizing advancements in machine learning and neural network architectures to enhance accuracy and efficiency in industrial quality control processes. This paper makes a notable contribution by addressing the early stage of research in automating quality control, where scientific publications on this topic are limited. Leveraging transfer learning from the pre-trained model, the study investigates the potential improvement of surface defect detection in a smaller dataset.

3 METHODOLOGY

This research focuses on the systematic approach which is adopted to increase efficiency and reduce errors in mechanical design. A well-curated dataset includes real-time images and synthesized 3D-CAD data to ensure a variety of representative training models. Leveraging the VGG 16 Imagenet algorithm within a Convolutional Neural Network (CNN), the system accurately categorizes components. The overall methodology is shown in Figure 2. Data augmentation techniques, rigorous testing, and a user-friendly Graphical User Interface (GUI) contribute to an effective system. These components enable manufacturers to gain data-driven insights for real-time productivity improvements and demonstrate the potential for future innovation in manufacturing.

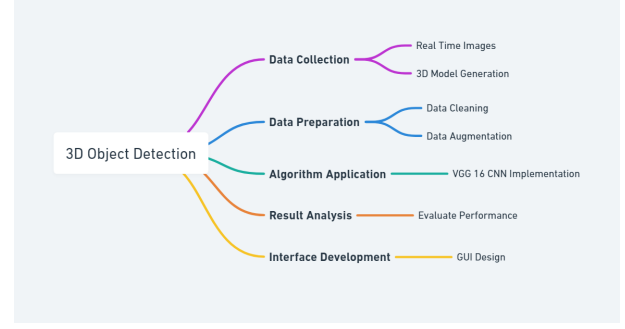


Figure 2: Overall Methodology

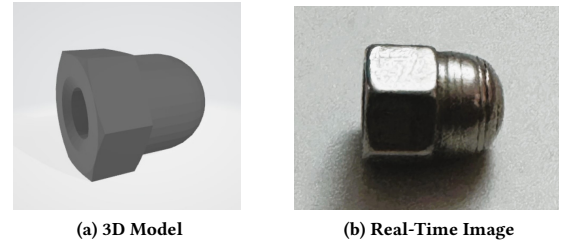


Figure 3: Images used for Hexagon Cap Nut

3.1 Data Collection

The dataset is meticulously curated, combining both synthetic 3D-CAD data and real-time photos. Figure 3 shows the images for the hexagon cap nut, both the 3D model and the real-time image that is used in the model. The purpose of the synthetic 3D-CAD data was to include controlled variations and improve the model's generalizability in various manufacturing conditions. Specific manufacturing parts were acquired, and in-depth photos were taken, to further enhance the dataset. This guarantees a thorough and varied depiction of manufacturing components at various stages of development.

3.2 System Architecture

3.2.1 CNN Selection. Our method's primary element is the incorporation of the VGG 16 ImageNet algorithm within a Convolutional Neural Network (CNN) architecture. We have strategically opted for this since this method has an established history of success in image classification tasks. The CNN has been specifically constructed to handle a crucial function, which is differentiating between different phases of the production process. Our network can make a substantial contribution to quality control procedures by evaluating the accuracy, consistency, and integrity of manufacturing components thanks to this refined focus. Additionally, the CNN has been fine-tuned to guarantee precise classification in a variety of circumstances, such as varying angles and different lighting scenarios. This flexibility guarantees our system's strong performance in actual manufacturing settings where lighting and perspectives might differ greatly.

3.2.2 Data Augmentation. To improve the model's generalization and to increase the diversity of the dataset, data augmentation techniques have been applied. Common augmentations include rotation, flipping, zooming, and slight translations. Rotation and scaling are used as data augmentation techniques to improve dataset heterogeneity. This step attempts to enhance generalisation and strengthen the model against variability found in the real world. This makes the model able to adapt to changes in manufacturing environments.

3.2.3 Graphical User Interface. To improve usability, a basic Graphical User Interface (GUI) is created. This GUI enhances accessibility and usability by enabling users to examine and analyse manufacturing components in a natural manner accessibility and usability. The GUI starts the data creation process when the user selects STL File. The resulting images are stored in the specified folder. Once the data is captured, the backend performs preprocessing to prepare the data for training. The GUI has a "Train" button, which starts the model training process. Users can monitor the training progress through the displayed accuracy loss graphs. In addition, a confusion matrix is provided to evaluate the performance of the model. After successful training, users have the option to save the trained model. We makes it possible to save the sample for future use. The GUI also facilitates the loading of previously saved models. It includes graphics such as accuracy plots, loss plots, and confusion matrices to provide users with a detailed overview of the model's performance.

3.3 Model Training

Training of the CNN involves using real-time images captured. The dataset includes images of three different parts, each captured from both top and front views. The pre-trained VGG 16 Imagenet model is leveraged for transfer learning. This approach accelerates the training process and enhances the model's ability to extract relevant features from manufacturing components. The extracted features are used to train a classification model. A fully connected layer is appended to the VGG16 output to map the features to the class labels. Figure 4 summarizes the VGG root system and transfer learning process. It illustrates the layers of a VGG network and shows how a pre-trained VGG network can be modified and optimized for the specified task. The model is trained using a suitable loss function, such as categorical cross-entropy.

$$\text{Loss} = - \sum_{i=1}^N \left[\frac{1}{\ln(2)} \int_{-\infty}^{\infty} \frac{\text{GT}_i}{\sigma_i \sqrt{2\pi}} \exp \left(-\frac{(\ln(\text{PP}_i) - \mu_i)^2}{2\sigma_i^2} \right) d(\ln(\text{PP}_i)) \right] \quad (1)$$

where:

GT : Ground Truth : Represents the true labels or actual class values.

PP : Predicted Probability : Stands for the probabilities estimated or predicted by the model.

N : Denotes the number of classes.

μ_i and σ_i : Signify the mean and standard deviation of the logarithm of estimated probabilities for class i .

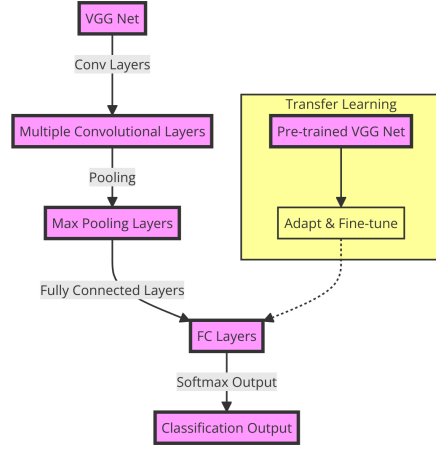


Figure 4: VGG Net Architecture and Transfer Learning Process

3.4 Evaluation and Validation

To assess the trained model's performance, the structural similarity index (SSIM) is computed. A strong performance in capturing structural similarities between expected and actual images is indicated by the achieved SSIM score of 0.86.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (2)$$

where μ is the mean, σ is the standard deviation, σ_{XY} is the covariance, and c_1, c_2 are constants.

To visually evaluate the model's capacity to capture fine features, edge detection is used. This stage offers more information about the network's capacity to identify characteristics that are essential for precise classification. CNN is put through thorough testing with real-world production scenario images. Variations in lighting conditions and field orientation are considered to confirm the reliability of the model under practical building conditions.

3.5 System Deployment

Our model is optimised to meet the particular requirements of real-time deployment, using resources effectively without compromising performance. Rather than exploring other CNN architectures, we intentionally narrowed down our selection to the VGG 16 ImageNet architecture since we are aware that it works well to meet the unique needs of our application that is focused on manufacturing environments. This focused strategy increases the viability of using our solution in manufacturing environments with limited resources. The generated confusion matrix indicated that the model does an effective job of classifying the dataset, but it also pointed out particular areas where the model could be refined to achieve even greater results. To confirm the reliability of the model, we not only evaluate its capacity to handle partial views and blind spots, but we also simulate real-world production scenarios. The assessment reaffirms our simplified CNN architecture's effectiveness in actual manufacturing environments.

3.6 Impact and Contribution

Once effectively implemented, the developed system offers valuable insight into manufacturing procedures. This research has significant implications, as demonstrated by our Convolutional Neural Network (CNN) that we constructed based on the VGG 16 ImageNet algorithm. It can recognise maturity levels across the manufactured parts with an accuracy rate of 93%. The project emphasizes the importance of implementing the VGG 16 Image-net algorithm in addition to advancing manufacturing processes. This establishes the foundation for upcoming advancements in material handling, process control, and overall manufacturing efficiency.

4 RESULTS

Image	True Class	Predicted Class
	Hexagoncap_front	Hexagoncap_front
	Theadbolt_front	Theadbolt_front

Table 1: Classification Results

In Table 1, we present the classification results for two different types of manufacturing components. It provides insight into how well the model recognizes specific components, showcasing the model's capacity to distinguish and classify various parts based on front-view images. The performance of the model is demonstrated, along with some of the ways it could be used to enhance manufacturing processes through precise part identification and classification. The first row depicts the front portion of a hexagonal cap, which our model correctly identifies as 'Hexagoncap_front' in both the True Class and Predicted Class. The second row depicts the front of a T Head Bolt, which is also classified by our model as 'Theadbolt_front' for both the true and predicted classes. These outcomes show how well the model can distinguish and categorize various manufacturing parts from frontal view images. Rather than showing raw counts, the figure 5 exhibits normalized version to show percentages of correct and incorrect predictions for each class. The confusion matrix indicates that the model does an effective job of classifying the dataset, but it also points out particular areas where the model could be refined to achieve even greater results.

5 CONCLUSIONS

5.1 Overview

In conclusion, this research describes a thorough method for automating the VGG 16 Imagenet-based Convolutional Neural Network (CNN) and 2D camera-based maturity level recognition across multiple manufacturing parts. Using systematic dataset, merging synthetic 3D and real data, and strategically deploying data augmentation, the framework is designed to improve overall manufacturing process efficiency, reduce errors, and enhance productivity.

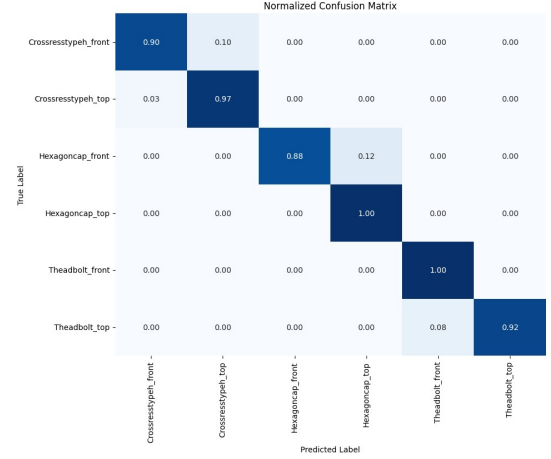


Figure 5: Normalized Confusion Matrix

The chosen VGG 16 ImageNet algorithm exhibits its efficacy in accurate classification tests and is easily incorporated into our CNN architecture. The superior performance of the model is ensured by rigorous training, which includes transfer learning, hyper parameter optimisation, and careful epoch management. The model's capacity to capture structural details is confirmed by the computed SSIM score of 0.86 and edge detection visualisations. Creating a Graphical User Interface (GUI) that is easy to use improves accessibility and makes the system useful for manufacturing processes.

5.2 Future Work

Future research initiatives might focus on expanding the dataset in order to improve the model's knowledge of real-world changes. The adaptability and classification accuracy of the system could be increased by investigating more sophisticated transfer learning strategies and refining the integration of 3D-CAD data. Furthermore, there is possibility of expanding the system's use to other sectors, such quality assurance and logistics. The system's relevance will be maintained in the face of changing environmental circumstances and industrial processes through continuous model monitoring and update methods.

ACKNOWLEDGMENTS

- We would like to extend our sincere thanks to **Professor Jan Schmitt**, who has offered his invaluable guidance and insights throughout the progression of this project. Professor Schmitt's mentorship provided us with direction and insight that deepened our trust in our subject matter, and inspired us to reach for the quality we have achieved herein.
- A special note of gratitude to **Mr. Eddi Miller**, who not only provided invaluable assistance to this project in its development but kept the myriad of documentation associated with such a long term work together with a grace and attention to detail we are truly appreciative for.

- Thanks to **Whimsical.com**, whose intuitive tool helped us to create information dense diagrams with ease, this versatility allowed our data to be presented in a engaging and clear manner, greatly adding to the quality of our project.
- Special thanks to **Overleaf.com** for giving a brilliant platform to write and have multiple collaborators work on this content in the LaTeX format. Overleaf was not only the easiest, but made the professional quality of our documents assured by its very interface and robust features.

REFERENCES

- [1] Georgia Andriosopoulou, Andreas Mastakouris, Dimosthenis Masouros, Panorios Benardos, George-Christopher Vosniakos, and Dimitrios Soudris. 2023. Defect Recognition in High-Pressure Die-Casting Parts Using Neural Networks and Transfer Learning. *Metals* 13, 6 (2023), 1104.
- [2] İlhan Aydin, Mehmet Karakose, G Ghazi Hamsin, Alisan Sarimaden, and Erhan Akin. 2017. A new object detection and classification method for quality control based on segmentation and geometric features. In *2017 International Artificial Intelligence and Data Processing Symposium (IDAP)*. IEEE, 1–6.
- [3] Carlos A Escobar and Ruben Morales-Menendez. 2018. Machine learning techniques for quality control in high conformance manufacturing environment. *Advances in Mechanical Engineering* 10, 2 (2018), 1687814018755519.
- [4] Yakov Frayman, Hong Zheng, and Saeid Nahavandi. 2006. Machine vision system for automatic inspection of surface defects in aluminum die casting. (2006).
- [5] Iker Pastor-López, Igor Santos, Jorge de-la Peña-Sordo, Mikel Salazar, Aitor Santamaria-Ibirika, and Pablo G Bringas. 2013. Collective classification for the detection of surface defects in automotive castings. In *2013 IEEE 8th Conference on Industrial Electronics and Applications (ICIEA)*. IEEE, 941–946.
- [6] Iker Pastor-López, Igor Santos, Aitor Santamaria-Ibirika, Mikel Salazar, Jorge De-la Peña-Sordo, and Pablo G Bringas. 2012. Machine-learning-based surface defect detection and categorisation in high-precision foundry. In *2012 7th IEEE Conference on Industrial Electronics and Applications (ICIEA)*. IEEE, 1359–1364.
- [7] Tian Wang, Yang Chen, Meina Qiao, and Hichem Snoussi. 2018. A fast and robust convolutional neural network-based defect detection model in product quality control. *The International Journal of Advanced Manufacturing Technology* 94 (2018), 3465–3471.
- [8] Zhang Xue-Wu, Ding Yan-Qiong, Lv Yan-Yun, Shi Ai-Ye, and Liang Rui-Yu. 2011. A vision inspection system for the surface defects of strongly reflected metal based on multi-class SVM. *Expert Systems with Applications* 38, 5 (2011), 5930–5939.